

SAYLÉ SIGARRETA

PH.D. CANDIDATE IN MATHEMATICS

CONTACT INFO



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RESEARCH INTERESTS

My research interests lie at the intersection of probability and combinatorics, with applications to graph theory. I am particularly interested in free probability and non-commutative probability. My interests in probability include limit theorems and random matrix theory, while in combinatorics I focus on set partitions and cumulants. I also work in graph theory, particularly spectral graph theory, graph energy, and random graphs.

EDUCATION

2023-2026 PhD in Mathematics, expected December 2026, Benemérita Universidad Autónoma de Puebla, México

Advisors: Octavio Arizmendi Echegaray and Hugo Adán Cruz Suárez.

2021-2023 Master of Science, magna cum laude, Benemérita Universidad Autónoma de Puebla, México

2016-2020 Bachelor of Science, Benemérita Universidad Autónoma de Puebla, México

SELECTED PUBLICATIONS

2026

- Arizmendi, O., George, D. M., & Sigarreta, S. Third Order Cumulants of Products. Annales Henri Poincaré. Forthcoming.
- Montes-y-Morales, M., Sigarreta, S., & Cruz-Suárez, H. Maximum augmented Zagreb index on polyomino chains. Applied Mathematics and Computation, 519, 129929.

2025

- Eslava, L., Sigarreta, S., & Siri-Jégousse, A. Limit Theorems for Randić Index for Erdős-Rényi Graphs. In Progress in Probability, 81, 165-178. Birkhäuser, Cham.

2024

- Arizmendi, O., & Sigarreta, S. The change of vertex energy when joining trees. Linear Algebra and its Applications, 687, 117-131.

RESEARCH TALKS

2026

- Invited speaker, Third-Order Cumulants: A Bridge Between Combinatorics and Statistics, International Congress on Mathematics and Its Applications.
- Invited speaker, Functional Equations for (ab) in Second-Order Free Probability, II French-Mexican Workshop in Probability.

2025

- Third-order cumulants of products, XV Symposium of Probability and Stochastic Processes.
- Energy change after coalescing, Interinstitutional Seminar on Random Matrices.
- On the third-order cumulants of products, Montreal-Guanajuato Workshop on Probability and Machine Learning.
- Asymptotic Behavior of the Randić Index in Erdős–Rényi Graphs, International Congress on Mathematics and Its Applications.

2024

- Energy change after fusion, Quantum Markov Semigroups and Channels
- Topological Indices: Exploring Random Structures, Seminar on Probability and Stochastic Processes, UNAM